Architecture decision record | A Nigerian payment processor

PAYMENTS GATEWAY

HYBRID MODERNISATION

Decision requested: Endorse ADR-0001 to ADR-0004, and ask the compliance team to decide ADR-0005 before the foundations phase starts.

Sakariyau Oseni | Version 1.0 | 18 October 2026 | 5 decisions: 4 accepted, 1 proposed

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EXECUTIVE SUMMARY

Move three products to AWS, keep the record in Lagos, and make sure every payment is delivered once and never lost.

PROBLEM

One on-premises table on every payment's path, a duplicate check that reads and then acts, and no rule for a vendor timeout.

RECOMMENDATION

Claim each payment once in DynamoDB, stream every change to Lagos through Amazon MSK, re-query unknown outcomes, keep state revenue on premises, run in Cape Town (proposed).

3M	PAYMENTS A DAY; 300 A SECOND AT PEAK
0	PAYMENTS DELIVERED TWICE, BY DESIGN
$2,690	A MONTH ON DEMAND; $0.03 PER 1,000 PAYMENTS
27/27	TESTS PASSED LOCALLY, IN CI AND ON AWS

DECISION REQUESTED Endorse ADR-0001 to ADR-0004, and ask the compliance team to decide ADR-0005 before the foundations phase starts.

FIVE PRESSURES

01 One transaction table on every payment's path

02 Duplicates from retries: USSD drops, terminal timeouts, double taps

03 Vendor timeouts leave the outcome unknown

04 One data centre, capacity bought for the peak

05 No AWS Region in Nigeria; payment records stay in Lagos

EIGHT REQUIREMENTS

REQ-01 VOLUME
3 million payments a day; 300 a second at peak

REQ-02 DUPLICATES
None delivered, ever

REQ-03 DURABILITY
No accepted payment lost (recovery point zero)

REQ-04 LATENCY
Gateway adds at most 300 ms at p95, vendor time excluded

REQ-05 AVAILABILITY
99.95% a month for the cloud tier

REQ-06 RECORD
99% of changes in Lagos within 5 s; none lost in a 7-day link outage

REQ-07 UNKNOWNS
Settled within 15 minutes, or passed to operations

REQ-08 RESIDENCY
Claims held 48 hours, events 7 days, nothing longer

SCOPE AND ASSUMPTIONS

IN

IN SCOPE

- Electricity, airtime and data, and pay-TV moved to AWS
- State revenue routed to Lagos, unchanged
- Syncing every change to the system of record
- Network, security, cost and migration order

OUT

OUT OF SCOPE

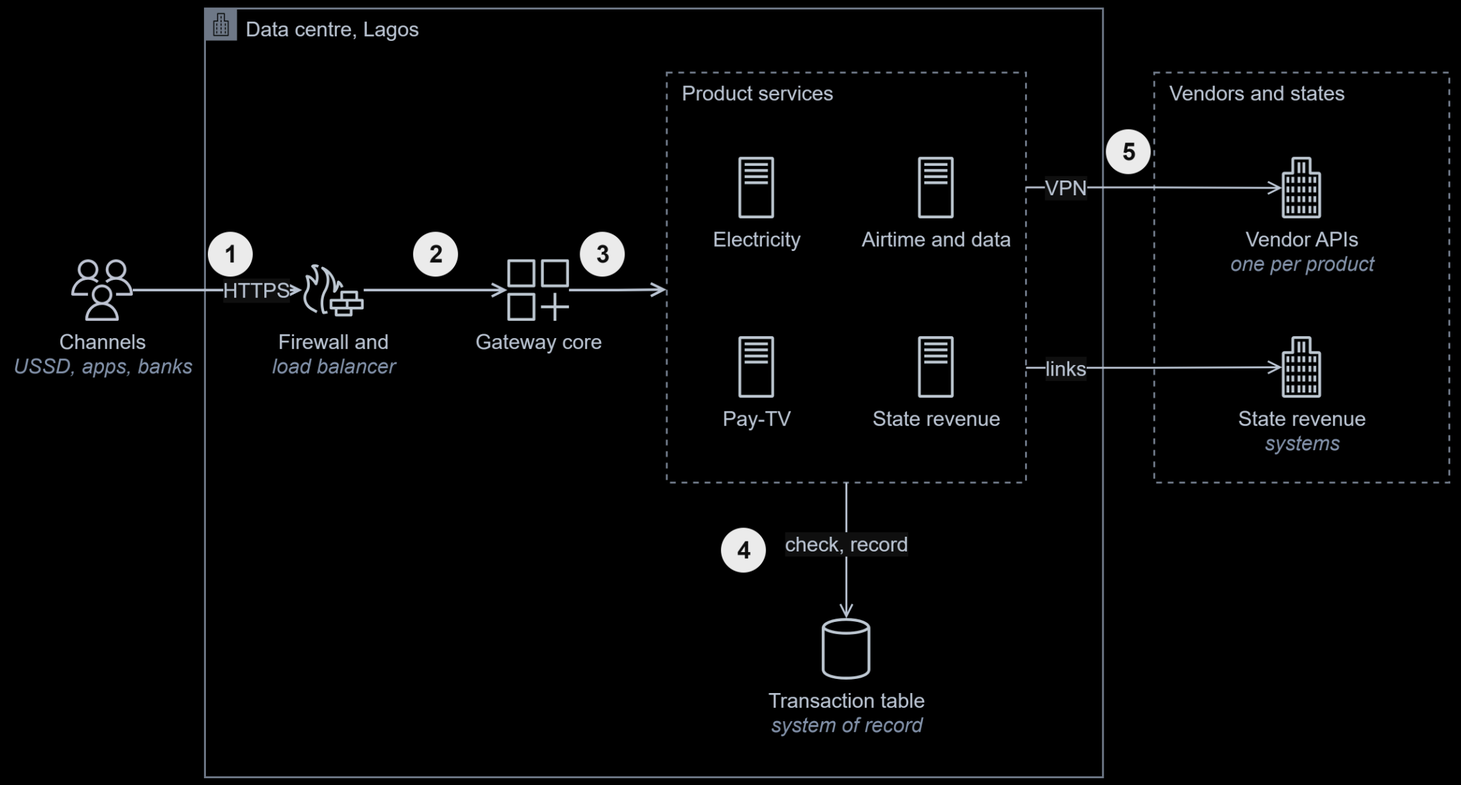
- Channels' own systems and card acquiring
- Settlement and reconciliation engines
- A second AWS Region

IF

ASSUMPTIONS

- Every vendor offers a status query and logs every request it accepts (to go into contracts)
- State integrations trust the data centre's addresses (to be confirmed by an inventory)
- Volumes and latency targets are mine: the original analysis left them blank
- Compliance decides where payment data may live; ADR-0005 waits on it

CURRENT STATE: ONE TABLE ON EVERY PAYMENT'S PATH



1 Channels reach the data centre over HTTPS

2 A firewall and load balancer pass requests to the gateway core

3 The core routes each payment to its product service

4 Every product checks the one table for duplicates, then records state there: two trips per payment

5 Product services call vendors over VPN; state revenue uses its own links

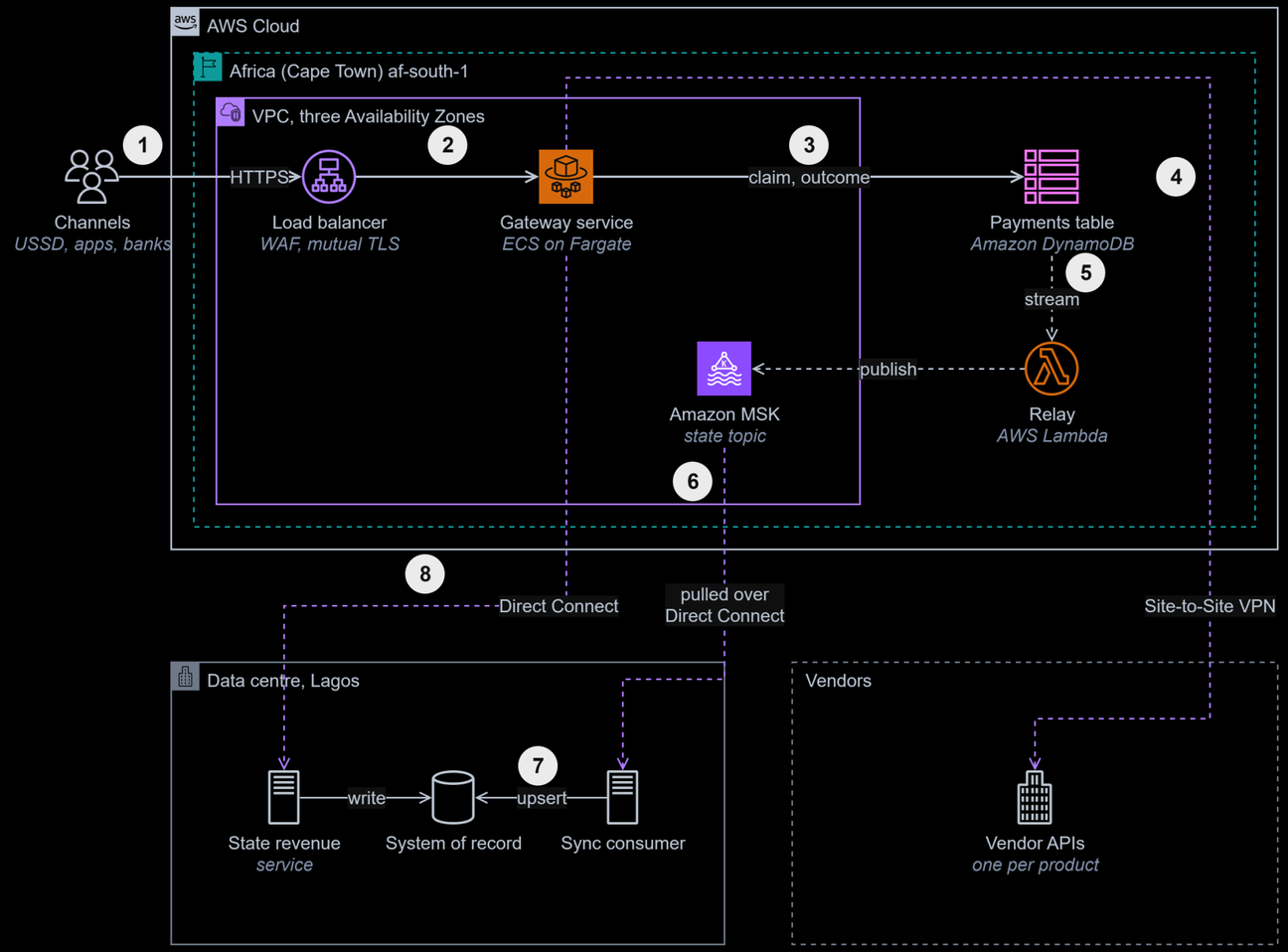
FIGURE 1 | CURRENT STATE, ONE DATA CENTRE IN LAGOS

CURRENT-STATE FINDINGS

1 CRITICAL | 3 HIGH

ID AND AREA	FINDING	SEVERITY	ADDRESSED BY
F1 Correctness	Duplicate check reads, then forwards in a separate step	CRITICAL	ADR-0001
F2 Correctness	No rule for a vendor timeout	HIGH	ADR-0004
F3 Scalability	One table on every payment's path, twice	HIGH	ADR-0001, ADR-0002
F4 Resilience	One data centre, capacity bought for the peak	HIGH	Target state
F5 Operations	Throughput and latency never measured	MEDIUM	REQ-01, REQ-04

TARGET STATE: THREE PRODUCTS IN AWS, THE RECORD IN LAGOS



1 Channels send HTTPS with a client certificate; WAF and the load balancer check it

2 The gateway service on ECS with Fargate validates and routes

3 One conditional write claims the payment; its outcome is recorded on the same item

4 The vendor is called over its own Site-to-Site VPN

5 DynamoDB Streams and the Lambda relay publish each change to Amazon MSK

6 The sync consumer in Lagos pulls the topic over Direct Connect

7 It writes the system of record, newer versions only

8 State revenue goes to Lagos over Direct Connect, as today

FIGURE 2 | TARGET STATE, CAPE TOWN AND LAGOS

A PAYMENT'S STATES

PENDING	Claimed; the vendor is being called
SUCCESS	Delivered; the token is stored for duplicates
FAILED	Declined, or no vendor record when the 15-minute window closes
UNKNOWN	Timed out after sending: re-queried after 30 s, 1, 2, 4 and 8 min
REVIEW	The vendor keeps erroring: operations and reconciliation settle it

FAILURE HANDLING

20 identical requests at once	One claim wins; all 20 get the same answer
The vendor answers too late	UNKNOWN, then SUCCESS from the re-query; one delivery
A task stops mid-payment	The stale PENDING claim is re-queried after 60 s
Direct Connect fails	Backup VPN; the three cloud products continue
The consumer is down	Changes wait on the topic up to 7 days, then catch up

DECISION LOG

ADR-0001	Claim each payment once with a DynamoDB conditional write	Accepted	REQ-02, REQ-03, REQ-04
ADR-0002	Sync the system of record through DynamoDB Streams and Amazon MSK	Accepted	REQ-03, REQ-06
ADR-0003	Keep state revenue payments on premises	Accepted	REQ-06
ADR-0004	Resolve unknown outcomes by re-querying, never by re-sending	Accepted	REQ-02, REQ-07
ADR-0005	Run the cloud tier in Cape Town, holding payment data for at most seven days	Proposed	REQ-05, REQ-08

ADR-0001

CLAIM EACH PAYMENT ONCE WITH A DYNAMODB CONDITIONAL WRITE

CONTEXT

Channels retry and sometimes send the same payment twice at once. Today's check reads the table, then forwards in a separate step, so two identical requests can both be sent.

Options weighed

- DynamoDB conditional write (chosen)
- Redis SET NX in ElastiCache
- Read the cache, then forward
- Synchronous check in Lagos

DECISION

ACCEPTED

We will claim each payment with one write that succeeds only if the reference is new or its 48-hour claim has expired.

Confidence	High
Requirements	REQ-02, REQ-03, REQ-04

Confirmation

20 identical requests at once produce one delivery (demo and AWS run)

CONSEQUENCES

Good

- Atomic: of any number of duplicates, one claim wins
- Durable once acknowledged, across Availability Zones
- Duplicates get the original answer back

Bad

- Single-digit milliseconds, not sub-millisecond
- Payment data held in the cloud for the window (ADR-0005)

ADR-0002

SYNC THE SYSTEM OF RECORD THROUGH DYNAMODB STREAMS AND AMAZON MSK

CONTEXT

The system of record stays in Lagos. Services that write the state and then publish it lose the change if they stop between the two writes.

Options weighed

- Streams, Lambda relay, MSK (chosen)
- Each service publishes to Kafka
- Kinesis Data Streams for DynamoDB
- Synchronous writes to Lagos

DECISION

ACCEPTED

We will write each change once to DynamoDB; the stream and a Lambda relay publish it to MSK, keyed by payment, and Lagos pulls it.

Confidence	Medium
Requirements	REQ-03, REQ-06

Confirmation

Consumer stopped: 5 payments arrive after restart; a second replay changes nothing

CONSEQUENCES

Good

- No dual write; order kept per payment
- Payments continue through a 7-day link outage
- The record's zone opens no inbound port

Bad

- MSK is the largest line: about $630 a month
- The record lags by seconds; 24-hour stream window to watch

ADR-0003

KEEP STATE REVENUE PAYMENTS ON PREMISES

CONTEXT

State revenue integrations are built one state at a time, over links that trust the data centre's addresses.

Options weighed

- Keep them in Lagos, routed over Direct Connect (chosen)
- Move them with the other products
- Split traffic before the gateway

DECISION

ACCEPTED

We will leave the integrations and their writes in Lagos; the cloud gateway routes state revenue there.

Confidence Medium

Requirements REQ-06

Confirmation

An inventory of each state's dependencies; a failover test over the backup VPN

CONSEQUENCES

Good

- No state has to change its side
- Revenue records are in the system of record at once

Bad

- Two duplicate checks
- Stops if Direct Connect and the backup VPN both fail

ADR-0004

RESOLVE UNKNOWN OUTCOMES BY RE-QUERYING, NEVER BY RE-SENDING

CONTEXT

A timeout leaves the gateway not knowing whether the vendor delivered. Re-sending can pay twice; failing can refund a delivered token.

Options weighed

- Re-query until the window closes (chosen)
- Re-send with the same reference
- Fail at once and reverse
- Leave it to end-of-day reconciliation

DECISION

ACCEPTED

We will record UNKNOWN, answer PENDING, and re-query for 15 minutes; never send the same payment twice.

Confidence High

Requirements REQ-02, REQ-07

Confirmation

A late vendor ends SUCCESS with one delivery; a lost request ends FAILED with none

CONSEQUENCES

Good

- No payment delivered twice
- Most unknowns settle within minutes

Bad

- Channels must handle PENDING
- Relies on vendors logging every request (contracts)

ADR-0005

RUN THE CLOUD TIER IN CAPE TOWN, HOLDING PAYMENT DATA FOR AT MOST SEVEN DAYS

CONTEXT

No AWS Region in Nigeria; the Lagos Local Zone has no DynamoDB, MSK or Fargate. Where payment data may live is for compliance to decide.

Options weighed

- af-south-1, 7-day retention (proposed)
- Lagos Local Zone, self-managed stores
- A European Region
- Stay on premises

DECISION

PROPOSED

We propose af-south-1, with claims expiring after 48 hours, events after 7 days, and the record kept in Lagos.

Confidence	Medium
Requirements	REQ-05, REQ-08

Confirmation

Written approval before production; latency test from Lagos; SCP and Config rule

CONSEQUENCES

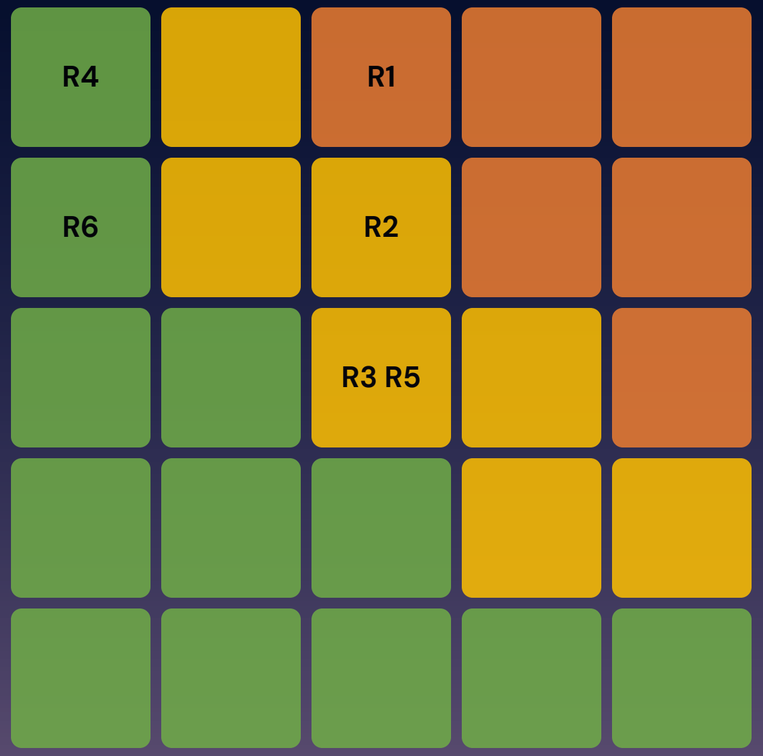
Good

- Managed services in three Availability Zones
- No payment record in the cloud beyond 7 days

Bad

- At least 96 ms of each payment is the distance
- Depends on compliance's written approval

RISKS AND MITIGATIONS



LIKELIHOOD 1 TO 5, LEFT TO RIGHT
IMPACT 1 TO 5, BOTTOM TO TOP
GREEN LOW 1 TO 7 | AMBER MODERATE 8 TO 14 | ORANGE HIGH 15 TO 25

ID	RISK AND MITIGATION	OWNER AND SCORE
R1	Compliance refuses data in Cape Town Lagos Local Zone fallback, self-managed stores	Compliance 15 · High
R2	A vendor cannot answer status queries Contract clause before cut-over; REVIEW path	Vendor management 12 · Moderate
R3	Added latency over 300 ms at p95 Load test from Lagos before each product moves	Architecture 9 · Moderate
R4	Relay broken for over 24 hours 5-minute lag alarm; daily reconciliation	Operations 5 · Low
R5	Team new to Kafka and DynamoDB Training; partner support for the first quarter	Engineering 9 · Moderate
R6	Direct Connect and backup VPN both fail Two Direct Connect ports; cloud keeps running	Network 4 · Low

MONTHLY COST, INDICATIVE

LINE ITEM AND BASIS	USD A MONTH
Amazon MSK · 3 kafka.m7g.large brokers, 300 GB	$631
AWS Direct Connect · 2 dedicated 1 Gbps ports	$460
AWS Transit Gateway · 8 attachments, 600 GB	$362
AWS Site-to-Site VPN · 6 vendors, 1 backup	$256
Amazon ECS on AWS Fargate · 6.5 vCPUs, 13 GB	$253
Route 53 Resolver endpoint · 2 network interfaces	$183
Interface VPC endpoints · 6 services in 3 zones	$173
Amazon DynamoDB · 180 million writes, PITR	$158
WAF, ALB, transfer, logs, KMS, Lambda	$218
TOTAL	$2,690

$2,690

A MONTH ON DEMAND IN AF-SOUTH-1

ASSUMPTIONS

- 3 MILLION PAYMENTS A DAY: $0.03 PER 1,000
- ON-DEMAND PRICES, AWS PRICE LIST API, 18 OCTOBER 2026
- CONNECTIVITY IS 47% OF THE BILL
- EXCLUDES SUPPORT, COLOCATION AND VENDORS' VPN ENDS

PROOF: DEMO AND LIVE RUN ON AWS

WHAT TESTING CAUGHT

01 A token-provider import that would have crashed on MSK
02 A replay test that assumed a fresh topic: replay restored 13 missed payments
03 A consumer that held its Kafka partitions for 45 s after SIGTERM

27 OF 27

tests passed against DynamoDB, the Lambda relay and Amazon MSK on AWS

75 / 0

relay invocations / errors in the live run

2.1 S

longest a change waited in the stream before it was published

$0.35

for the two-hour run, then all 42 resources destroyed

ROADMAP: ONE PRODUCT AT A TIME

0

PHASE 0 FOUNDATIONS

Compliance decision; landing zone; Direct Connect, VPNs, Transit Gateway.

Exit: ADR-0005 decided; links tested.

1 TO 3

PHASES 1 TO 3 ONE PRODUCT AT A TIME

Phase 1 Pay-TV through the cloud gateway. Exit: a month end with REQ-01 to REQ-07 met.

Phase 2 Airtime and data. Exit: load test at 300 a second.

Phase 3 Electricity. Exit: vendor status queries confirmed.

4

PHASE 4 RETIRE THE FALLBACK

Retire the on-premises fallback.

Exit: all three products through a month end.

NEXT STEPS

DECISION REQUESTED

Endorse ADR-0001 to ADR-0004, and ask the compliance team to decide ADR-0005 before the foundations phase starts.

01 COMPLIANCE

Decides ADR-0005 and the data classification

02 VENDOR MANAGEMENT

Adds status-query and request-logging clauses

03 NETWORK

Orders two Direct Connect ports in Lagos

04 ENGINEERING

Starts Kafka and DynamoDB training

EVERY PAYMENT DELIVERED ONCE, AND NEVER LOST.

SAKARIYAU OSENI

THANK YOU

github.com/ooseni/TESA-Cloud-Engineering